INCH-POUND

MIL-C-24640/14C
26 June 1995
SUPERSEDING
MIL-C-24640/14B
31 July 1989

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI-CONDUCTOR, AWG 22, TYPE 1XSOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Watertight)

FIRST - 7 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. AWG size and dimensions in accordance with Table I.

SECOND - Insulation material and basic construction requirements shall be in accordance with MIL-W-81044/12 and tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.

THIRD - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90% determined in accordance with MIL-C-27500.

FOURTH - Shield insulation of two polyester tapes or polyester tape and extruded jacket. Identification codes applied using Method 2.

FIFTH - The required number of shielded components shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

SIXTH - Binder tape applied helically with overlap.

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SEVENTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface transfer Impedance requirements as listed herein.

EIGHTH - Binder tape applied helically with overlap.

NINTH - Crosslinked polyolefin jacket. (See Table II for thickness).

* TABLE I. Component wire details

Watertight conductor			Finished wire			
Wire size (AWG)	Stranding (no. x AWG)	Diameter nominal (inches)	Diameter (inches)	Test load life cycle accelerated aging tests (lbs. + 0.3%)	Low temp. at -40° C.	
					Mandrel diameter (inches)	Test load (lbs)
22	7 x 30	0.030	0.047 ± 0.003	0.190	0.75	1.0

TABLE II. Construction details.

PIN M24640/14	Type number	Number of component wires	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UO	1XSOW-2	2	0.173	0.034	0.292	0.314	85.0
-02UO	1XSOW-14	14	.362	.041	.470	.506	280.0
-03UO	1XSOW-20	20	.429	.044	.542	.584	330.0
-04UO	1XSOW-30	30	.520	.044	.614	.662	429.0

REQUIREMENTS:

Qualification required.
Construction details: See table I and II.

Component wire:

For the component wire the following additional tests shall be performed in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 135°C (275°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 250± 2°C (482± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)

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WIRE, RATINGS AND ADDITIONAL REQUIREMENTS (continued):

Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Identification, striping or banding durability: 125 cycles (250 strokes) (min), 500 gram weight.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 Khz spark test 5.5KV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 120 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.63
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	Not applicable
Conductor to shield	1000
Component Shield to Component Shield	200
Component Shield to overall shield	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield)	100
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	120
Capacitance unbalance - percent, maximum	Not applicable

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	Characteristic impedance - at 1 MHz, ohms	13 min, 27 max
	Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
	Weight	No failure
	Group B:	
	Cold bending	No failure
	Drip	No failure
*	Cross link proof test (percent maximum)	
	Jacket	50
	Tear - pounds per inch thickness, minimum	35
	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
	Gas flame (1 hour)	Not applicable
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged, minimum	50
*	Permanence of printing (conductor) - cycles, minimum	25
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - conformance to material, construction and coverage	Not applicable
	Component Shield - conformance to material, construction and coverage	No failure
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	No failure
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D	
	Flame propagation	No Failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Aging and compatibility (cable)	No failure
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
	Component wire	No failure

Unit ordering length: 500 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review Activities:
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Preparing Activity:
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Agent:
DLA-IS

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